Vaibhav Raheja

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PROFESSIONAL SUMMARY

Robotics Software Engineer with 2+ years building and deploying autonomous platforms across hardware-software architecture, multi-sensor perception, and production field operations. Strong across ROS2-based autonomy, multi-sensor fusion (LiDAR, thermal, PTZ), and full-stack robotics infrastructure, with field-proven results achieving 90% uptime and 40% improvement in fault detection rates.

WORK EXPERIENCE

TerraWise Solutions, Inc, *Founding Engineer* — Mar 2026 – Present | San Francisco, CA

- Leading the EarthSense-to-TerraWise platform migration, porting the full Solarbot stack from a ROS1/2 hybrid on NUC+Pi to a unified ROS2 Humble stack on a single NVIDIA Jetson Orin.
- Writing ROS2 Python drivers from scratch for ODrive (4WD skid-steer via CAN bus) and ZED-F9P GPS with NTRIP RTK correction.
- Building a unified safety watchdog triggering emergency-stops across battery, temperature, and GPS loss events, and a separate LiDAR-based obstacle stop monitor.

EarthSense, Inc, *Robotics Engineer / Project Lead* — Aug 2024 – Mar 2026 | Champaign, USA

- Led a cross-functional Solarbot team of 8+ engineers to deliver an autonomous solar panel inspection system across 3+ multi-site robot units.
- Redesigned the C++ motion controller safety state machine to detect zero-radius turns, enforce a full-stop intermediary between drive modes, and add configurable RPM ramp up/down, fixing misdetection bugs and hardening stop logic for field safety.
- Shipped 35+ PRs across the Python/C++ stack spanning motor control, per-motor RPM/current telemetry, camera bringup, and ARM64/Jetson platform deployment, all validated on a HIL rig I designed and built.
- Built a low-latency teleoperation system using AWS Kinesis WebRTC (UDP-based video transport) and WebSocket over TCP for command/control, enabling real-time remote monitoring of field robots.
- Integrated multi-sensor perception stacks (Hesai/Unitree LiDAR, PTZ cameras, Seek thermal, RealSense) into production via ROS2 and Docker, stress-tested across solar farm deployments in 43–46°C conditions.

Intelligent Motion Laboratory, *Robotics Research Developer* — Aug 2023 – Dec 2023 | Champaign, USA

- Achieved sub-2mm tracking accuracy for autonomous robotic eye examinations using MediaPipe FaceMesh and ZED depth cameras on UR5.
- Designed a custom camera mount in Fusion 360 for the UR5, improving head tracking precision by 20%.

All India Institute of Medical Sciences (AIIMS) Hospital, *Robotics Research Developer, NMIMS* — Feb 2021 – May 2023 | Mumbai, India

- Engineered a robot-assisted intubation system on xArm5 with HOTAS teleoperation and real-time camera feedback for critical care.
- Fabricated a custom catheter end-effector with integrated miniature camera for real-time airway visualization.

SKILLS

Programming: Python, C++, ROS/ROS2, OpenCV, PyTorch, Machine Learning (ML), CNNs
Robotics & Perception: LiDAR Integration (Hesai), Multi-Sensor Fusion, Camera Calibration, Path Planning, Motion Planning, Control Algorithms, Reinforcement Learning, Gazebo, SLAM, Docker
Networking & Protocols: TCP/IP, UDP, WebSocket, WebRTC, CAN, Serial Communication
Tools: Linux, Git, Fusion 360, CAD, Arduino, Raspberry Pi, 3D Printing

PROJECTS

Benchmarking Control Algorithms for Unitree Go1 Robot — *Python, ISAAC Sim, Reinforcement Learning | 2024*

- Built an ISAAC Sim benchmarking framework for the Unitree Go1, comparing factory vs. RL-based locomotion across velocity tracking, stability, and terrain adaptability.
- Verified real-world RL controller performance by deploying "Walk These Ways" on a physical Go1 across grass, gravel, and inclines.

GRAIC Autonomous Racing — *Python, Hybrid A*, Path Planning, PD Control, Vehicle Dynamics | 2023*

- Achieved 40.8% lap time improvement on the Shanghai circuit using a Hybrid A* path planner tuned to vehicle kinematics and non-holonomic dynamics.
- Benchmarked three algorithms (Hybrid A*, Dynamic Programming, Spline Interpolation) and tuned a PD controller for steering and velocity.

Intelligent Ground Vehicle Competition (IGVC), Team DARVIN — *Python, ROS, OpenCV, ODrive, Velodyne LiDAR, Gazebo | 2019–2023*

- Served as Co-Captain, securing 2nd in the Cyber Challenge (2023) and 3rd in the Auto-Nav Challenge (2022) against 15+ international university teams.
- Developed SOCRATES 2.0 on ROS with Velodyne LiDAR avoidance, OpenCV lane detection, GPS navigation, and ODrive S1 motor control, Gazebo-validated and field-deployed.

EDUCATION

University of Illinois Urbana-Champaign — Aug 2023 – Dec 2024 | Champaign, USA
Master of Engineering in Autonomy and Robotics
Mukesh Patel School of Technology Management & Engineering — Jul 2019 – Jun 2023 | Mumbai, India
Bachelor of Technology in Computer Engineering

PUBLICATIONS

V. Raheja, V. Shah, M. Shetty, and P. Patel, "Multi-Disease Prediction System using Machine Learning," *2022 International Conference on Futuristic Technologies (INCOFT)*, Belgaum, India, 2022, pp. 1–6. doi: 10.1109/INCOFT55651.2022.10094382